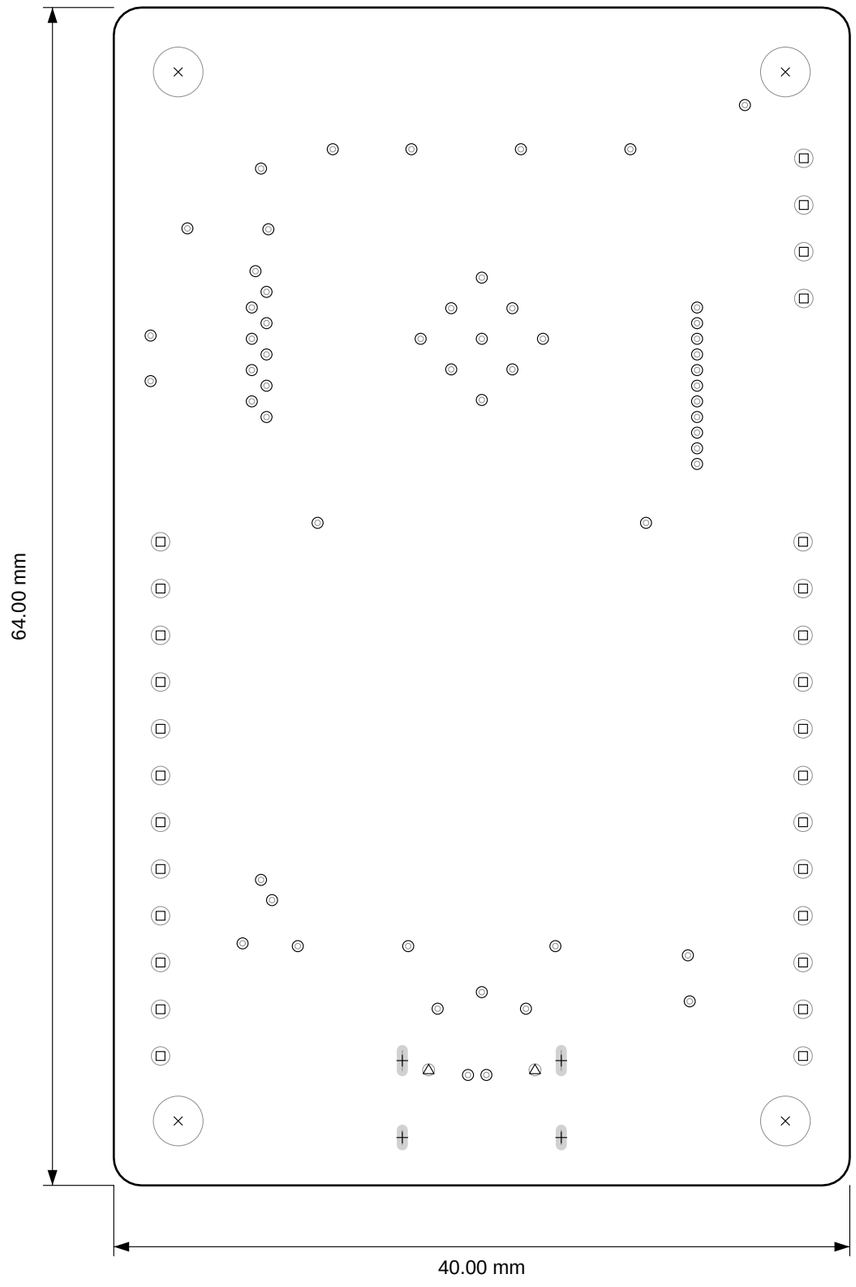
ESP32-S31 dev board — fabrication drawing

Viewed from the top. Scale 3:1. All dimensions in mm.



Drill table (93 holes, finished sizes)

Sym	Tool	Ø mm	Plated	Kind	Use	Count
◯	T1	0.300	yes	hole	via	55
┼	T2	0.600	yes	slot	component	4
□	T3	1.005	yes	hole	component	28
△	T4	0.650	no	hole	mechanical	2
×	T5	2.700	no	hole	mechanical	4

Stackup (4 copper layers, 1.74 mm over copper)

Layer	Material	mm
Top legend	white ink	—
Top mask	green solder mask	0.012
Top copper	35 µm Cu	0.035
Prepreg 1	FR-4 2116	0.200
Inner 1 copper	35 µm Cu	0.035
Core 1	FR-4 7628	1.200
Inner 2 copper	35 µm Cu	0.035
Prepreg 2	FR-4 2116	0.200
Bottom copper	35 µm Cu	0.035
Bottom mask	green solder mask	0.012
Bottom legend	white ink	—

Notes

1. Nominal board thickness 1.74 over the outer copper, solder mask and legend excluded; 4 copper layers.
2. Solder mask 12 µm thick over copper and 20 µm over laminate on each masked side.
3. Copper thickness top to bottom: 35 µm, 35 µm, 35 µm, 35 µm.
4. Minimum track width 0.09, minimum copper gap 0.09.
5. Minimum finished drill 0.15, via hole to copper 0.2, plated hole to copper 0.35, unplated hole to copper 0.2.
6. Minimum annular ring: vias 0.05, plated holes 0.2, copper round unplated holes 0.45.
7. Solder mask: green gloss, expansion 0, minimum web 0.1.
8. Legend: white ink, minimum line 0.15, minimum text height 1.
9. Surface finish: ENIG.
10. Dimensions in millimetres. Hole sizes are finished sizes; the drill file lists finished sizes. The fabricator does not publish how it reads drill files: confirm with it.
11. Waived by the designer, please confirm in DFM review: J1 pad 17: hole is 0.177 mm from J1 pad A1B12 on GND; needs 0.2 mm (JLCPCB 4-layer 1 oz: pad to unplated hole clearance). Reason: Manufacturer's recommended layout: GCT USB4105 drawing rev B4; confirm with the fab's DFM.
12. Waived by the designer, please confirm in DFM review: J1 pad 18: hole is 0.175 mm from J1 pad B1A12 on GND; needs 0.2 mm (JLCPCB 4-layer 1 oz: pad to unplated hole clearance). Reason: Manufacturer's recommended layout: GCT USB4105 drawing rev B4; confirm with the fab's DFM.
13. Generated by Sparksmith; the Gerber job file lists every layer.

ESP32-S31 dev board

Drawing	Fabrication drawing, sheet 1 of 1
Rules	JLCPCB 4-layer 1 oz (from JLCPCB, PCB Manufacturing & Assembly Capabiliti…
Board	40.00 × 64.00 mm, 4 layers, 1.74 mm
Finish	ENIG
Mask / legend	green / white
Generator	Sparksmith